Daniel Coblentz

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EDUCATION

Hood College — Frederick, MD — Expected May 2026
B.S. Computer Science | Minor: Mathematics — GPA:3.8/4.0 (Major GPA: 3.95)

Honors & Awards: Trailblazer Scholar (merit), Dean's List ×4, Volpe Scholar (top 2%), Christine P. Tischer Scholar (top 2%), Upsilon Pi Epsilon Honor Society (top 5%)
Related Coursework: Data Structures & Algorithms, Advanced Data Structures, Operating Systems, Object-Oriented Programming I/II, Database Management Systems, Machine Learning, Artificial Intelligence, Discrete Mathematics

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, SQL (PostgreSQL, MySQL), JavaScript, TypeScript, R, Bash
Frameworks & Tools: Spring Boot, React, Node.js, Docker, Git, GitHub Actions, Postman, CI/CD, API Design
Cloud & Deployment: AWS (EC2, ECS, Lambda), Google Cloud Platform, Heroku, Vercel

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern - Lawrence Berkeley National Laboratory — **June 2025 – Present**

- Engineered a multimodal ML pipeline using PyTorch and Transformers to fuse structured EHR data with clinical text, achieving 88% AUC in phenotype prediction and advancing healthcare data analysis.
- Applied contrastive learning (InfoNCE loss, cosine similarity) to align heterogeneous embeddings, improving cross-modal representation quality and model interpretability.
- Automated large-scale data ingestion, feature extraction, and UMAP-based dimensionality reduction for the MIMIC-IV dataset, accelerating experimentation and reproducibility.
- Optimized training performance and reproducibility through systematic hyperparameter tuning and grid search over 64 configurations, enhancing model accuracy and stability across validation folds.

Contract Software Engineer – Celebrations Catering — **September 2024 – May 2025**

- Designed and implemented a microservices-based scheduling system using Spring Boot and React to manage employee availability, prevent shift conflicts, and automate scheduling workflows.
- Refactored SQL queries and implemented caching, cutting database latency by 73% and improving scalability under concurrent load.
- Developed and tested RESTful APIs for shift management with JUnit and React Testing Library, achieving 92% test coverage and automated deployments via GitHub Actions CI/CD.

Data Analyst Intern – International Help — **May 2024 – August 2024**

- Developed automated Python pipelines to clean and validate survey data, reducing processing time by 10%.
- Analyzed user engagement trends with SQL and Pandas, increasing survey participation by 20%.
- Integrated survey data into internal dashboards, removing manual reporting, and simplifying updates.

PROJECTS

Book Recommendation Platform | Java · Spring Boot · React · PostgreSQL · Docker

- Built a full-stack recommendation engine integrating collaborative and content-based filtering, enabling personalized book discovery and improving user engagement.
- Designed secure RESTful APIs for authentication, ratings, and book management with JWT-based access control and role-based authorization.
- Applied query optimization and caching with Redis to accelerate recommendation delivery by 35%.

AI Code Review Dashboard | Java · Spring Boot · React · PostgreSQL · Docker · EC2 · ECS

- Engineered a real-time analytics dashboard to process 10K+ daily GitHub commits, reducing average code review latency by 45% through automated insights.
- Integrated CodeBERT and OpenAI APIs for intelligent code summarization and error detection.
- Deployed containerized microservices on AWS ECS with automated pipelines (GitHub Actions + CI/CD), maintaining 99.9% uptime and updates across 50 + repositories.